

Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 1: Large Numbers Around Us

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	1 — Large Numbers Around Us
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Every distractor is a number you reach by a real slip — a one-lakh-for-one-million swap, a comma shifted one period, place value confused with face value, rounding the wrong direction, or a digit miscounted by one. Do not write on this paper — mark answers on your answer sheet.

1. A 6-digit number has 8 in its lakhs place, 8 in its thousands place, and 0 in every other place. What is the number?

- (A) 88,000 (B) 8,00,800
(C) 8,08,000 (D) 80,08,000

2. In the number 4,86,30,517, add the digit in the ten-lakhs place to the digit in the lakhs place. The sum is:

- (A) 10 (B) 14
(C) 12 (D) 9

3. Rewrite the Indian number 6,07,00,050 in the International comma system.

- (A) 6,07,00,050 (B) 607,000,50
(C) 60,70,0050 (D) 60,700,050

4. In 9,30,80,704 the place value of the leftmost digit exceeds the place value of the digit 8 by:

- (A) 1 (B) 8,92,00,000
(C) 8,20,00,000 (D) 8,99,20,000

5. How many ten-thousands are there in 3 crore?

- (A) 300 (B) 3,000
(C) 30,000 (D) 3,00,000

6. The number name 'four crore four lakh forty thousand four' in figures is:

- (A) 4,44,40,004 (B) 4,04,04,004
(C) 44,40,004 (D) 4,04,40,004

7. A measurement is recorded as 250 million. Written in the Indian system with commas it is:

- (A) 2,50,00,000 (B) 25,00,000
(C) 2,50,00,00,000 (D) 25,00,00,000

8. Using each of the digits 0, 2, 5, 7, 9 exactly once, the largest 5-digit number that is even is:

- (A) 97,250 (B) 97,520
(C) 97,502 (D) 75,920

9. In 5,60,03,008, the difference between the place value and the face value of the digit 6 is:

- (A) 6 (B) 59,99,994
(C) 59,99,940 (D) 60,00,000

10. Reading right to left, the Indian system places commas after 3, then 2, then 2 digits. In the 9-digit number 123456789 the leftmost comma falls so the number reads:

- (A) 12,34,56,789 (B) 123,456,789
(C) 1,23,45,6789 (D) 1,2345,6789

11. Three numbers are 7,09,000, 7,90,000 and 7,00,900. Their correct DECREASING order is:

- (A) $7,09,000 > 7,90,000 > 7,00,900$ (B) $7,00,900 > 7,09,000 > 7,90,000$
(C) $7,90,000 > 7,00,900 > 7,09,000$ (D) $7,90,000 > 7,09,000 > 7,00,900$

12. Round 7,49,962 first to the nearest hundred, then round that result to the nearest thousand. The final figure is:

- (A) 7,49,000 (B) 7,50,000... 7,49,900
(C) 7,50,000 (D) 8,00,000

13. Estimate $4,12,000 \times 19$ by rounding the second factor to the nearest ten, then state the estimate.

- (A) 82,40,000 (B) 8,24,000
(C) 4,12,019 (D) 78,28,000

14. The expanded form $3 \times 1,00,00,000 + 5 \times 1,00,000 + 7 \times 1,000 + 2 \times 10$ represents:

- (A) 3,57,200 (B) 30,50,07,020
(C) 3,05,07,020 (D) 3,50,07,020

15. Without multiplying it out exactly, how many digits does the product $9,999 \times 9,999$ have?

- (A) 7 (B) 9
(C) 8 (D) 8 or 9

16. The number 'five crore five thousand five' in figures is:

- (A) 5,00,05,005 (B) 5,05,005
(C) 55,005 (D) 5,00,50,005

17. A publisher prints 4,800 copies of a book every day. Rounded to the nearest lakh, the number of copies printed in 30 days is:

- (A) 10,00,000 (B) 1,00,000
(C) 0 (D) 2,00,000

18. A relief train carries 1,250 sacks of grain, and 9 such trains run. To be sure of having enough sacks ready in the depot, the most sensible round figure to stock is:

- (A) round down to 11,000 (B) exactly 11,250
(C) round up to 12,000 (D) round down to 11,200

19. Counting at one number per second, about how long does it take to count to one lakh?

- (A) about an hour (B) a little over one day
(C) about one week (D) about one month

20. A district has 18 lakh people and the country it belongs to has 144 crore people. The district holds what fraction of the country's people?

- (A) $1/8$ (B) $1/80$
(C) $1/8000$ (D) $1/800$

21. The number 90,90,090 read in words in the Indian system is:

- (A) ninety lakh ninety thousand ninety (B) nine crore nine lakh ninety
(C) ninety lakh nine thousand ninety (D) ninety-nine lakh ninety

22. How many lakhs equal 12 crore?

- (A) 120 (B) 12,000
(C) 1,200 (D) 12

23. In the International system, 408,090,007 read in words is:

- (A) four hundred eight million ninety thousand seven
(B) four hundred eighty million nine thousand seven
(C) forty crore eighty lakh ninety thousand seven
(D) four hundred eight million nine hundred thousand seven

24. A scientist writes a distance as 2 crore 5 lakh kilometres. A foreign journal must print it in the International system with commas. It should read:

(A) 2,05,00,000

(B) 2,500,000

(C) 20,500,000

(D) 205,000,00

25. A digit has place value 30,00,000 in some number. By what factor is its place value larger than the place value of a digit sitting two places to its right (assuming the same face value)?

(A) 10

(B) 2

(C) 100

(D) 1,000

26. Which of these is the largest number?

(A) 79 million

(B) 8 crore

(C) 7,95,00,000

(D) 790 lakh

27. Round 5,49,52,000 to the nearest crore. The result is:

(A) 5,00,00,000

(B) 6,00,00,000

(C) 5,50,00,000

(D) 5,49,00,000

28. Estimate the sum $3,87,000 + 6,12,000 + 2,49,000$ by rounding each to the nearest lakh first. The estimate is:

(A) 13,00,000

(B) 11,00,000

(C) 12,00,000

(D) 12,48,000

29. How many digits does the product $25,000 \times 8,000$ have?

(A) 8

(B) 7

(C) 10

(D) 9

30. Which number, written in International figures, equals 'fifty million five hundred thousand fifty'?

(A) 50,005,050

(B) 50,500,050

(C) 50,500,500

(D) 500,500,050

31. A charity collected ₹2 crore 40 lakh and shared it equally among 16 villages. Each village received:

(A) ₹1,50,000

(B) ₹1,50,00,000

(C) ₹15,000

(D) ₹15,00,000

32. A 6-digit number reads the same forwards and backwards (a palindrome). Its lakhs digit is 4 and its tens digit is 7. What is its thousands digit, given the number is the SMALLEST such palindrome?

(A) 0

(B) 7

(C) 4

(D) 1

33. Compare $60 \text{ lakh} \times 5$ with 3 crore. Which statement is true?

(A) they are equal

(B) $60 \text{ lakh} \times 5$ is larger

(C) 3 crore is larger

(D) 3 crore is ten times larger

34. The same physical count is written 7,030,500 in the International system. In the Indian system with commas it is:

- (A) 7,03,05,00 (B) 70,30,50,0
(C) 7,03,0500 (D) 70,30,500

35. Written with Indian commas, the 10-digit number 1234567890 becomes:

- (A) 1,23,45,67,890 (B) 12,34,56,7890
(C) 1,234,56,7890 (D) 123,456,7890

36. In 8,07,60,400, the digit with the largest place value and the digit with face value 6 have place values that add to:

- (A) 8,00,60,000 (B) 8,60,00,000
(C) 14 (D) 8,00,06,000

37. Arrange in INCREASING order: 8.5 lakh, 0.9 million, 8,40,000, 95 ten-thousands.

- (A) 8.5 lakh < 8,40,000 < 0.9 million < 95 ten-thousands
(B) 8,40,000 < 0.9 million < 8.5 lakh < 95 ten-thousands
(C) 8,40,000 < 8.5 lakh < 0.9 million < 95 ten-thousands
(D) 95 ten-thousands < 0.9 million < 8.5 lakh < 8,40,000

38. Round 6,95,000 to the nearest ten-thousand, then to the nearest lakh. The two results are:

- (A) 7,00,000 and 7,00,000 (B) 6,90,000 and 6,00,000
(C) 7,00,000 and 6,00,000 (D) 6,90,000 and 7,00,000

39. A water plant pumps 3,75,000 litres per hour. About how many litres does it pump in one full day (round the hourly rate to the nearest lakh first)?

- (A) about 90,00,000 (B) about 9,00,000
(C) about 9,00,00,000 (D) about 96,00,000

40. Which is the correct expanded form of 7,06,500?

- (A) $7 \times 10,00,000 + 6 \times 1,000 + 5 \times 100$ (B) $7 \times 1,00,000 + 6 \times 10,000 + 5 \times 100$
(C) $7 \times 1,00,000 + 6 \times 100 + 5 \times 10$
(D) $7 \times 1,00,000 + 0 \times 10,000 + 6 \times 1,000 + 5 \times 100 + 0 \times 10 + 0 \times 1$

41. How many digits does the product of the smallest 4-digit number and the largest 4-digit number have?

- (A) 8 (B) 7
(C) 16 (D) 6

42. The number 'one arab two crore three lakh' in figures is:

- (A) 1,23,00,000 (B) 1,02,03,000
(C) 1,02,03,00,000 (D) 12,03,00,000

- 43.** A stadium holds 55,000 spectators. Over a season it is filled for 18 matches. Rounded to the nearest lakh, the total attendance is:
- (A) 9,00,000 (B) 10,00,000
(C) 1,00,000 (D) 1,00,00,000
- 44.** A printing error wrote 4,50,000 as 45,00,000. The wrong figure is too large by a factor of:
- (A) 100 (B) 9
(C) 10 (D) 1,000
- 45.** Two cities have populations 38 lakh and 0.6 crore. How much larger is the second city's population?
- (A) 22 thousand (B) 2.2 crore
(C) 22 lakh (D) 58 lakh
- 46.** At a counting rate of one number every 2 seconds, about how long does it take to count to one lakh?
- (A) about 1 day (B) about half a day
(C) about 1 week (D) a little over 2 days
- 47.** The same number is 1,209,000,034 in the International system. The number of digits before the FIRST Indian comma (reading right to left) is:
- (A) 2 (B) 3
(C) 4 (D) 1
- 48.** Arrange in DECREASING order: 9 crore, 88 million, 9,10,00,000, 900 lakh.
- (A) 9 crore > 9,10,00,000 > 900 lakh > 88 million
(B) 88 million > 9,10,00,000 > 9 crore > 900 lakh
(C) 900 lakh > 9,10,00,000 > 9 crore > 88 million
(D) 9,10,00,000 > 9 crore = 900 lakh > 88 million
- 49.** Round 9,99,500 to the nearest thousand, then to the nearest lakh. The two results are:
- (A) 10,00,000 and 10,00,000 (B) 9,99,000 and 10,00,000
(C) 10,00,000 and 9,00,000 (D) 9,99,000 and 9,00,000
- 50.** Estimate $5,000 \times 1,985$ by rounding the second factor to the nearest hundred, then state the estimate.
- (A) 99,25,000 (B) 10,00,000
(C) 1,00,00,000 (D) 1,00,000
- 51.** Which expanded form equals 80,07,060?
- (A) $8 \times 1,00,000 + 7 \times 1,000 + 6 \times 10$
(B) $8 \times 10,00,000 + 0 \times 1,00,000 + 0 \times 10,000 + 7 \times 1,000 + 0 \times 100 + 6 \times 10 + 0 \times 1$
(C) $8 \times 10,00,000 + 7 \times 10,000 + 6 \times 100$ (D) $8 \times 10,00,000 + 7 \times 100 + 6 \times 10$

52. How many digits does the product $3,00,000 \times 3,00,000$ have?

- (A) 11 (B) 10
(C) 12 (D) 6

53. A reservoir held 1 crore litres. First 35,80,000 litres drained out, then 12,50,000 litres were pumped back in. The litres now in the reservoir are:

- (A) 64,20,000 (B) 76,70,000
(C) 47,30,000 (D) 1,18,30,000

54. A number rounds to 50,00,000 when rounded to the nearest ten-lakh. Which of these CANNOT be that number?

- (A) 45,30,000 (B) 54,90,000
(C) 50,40,000 (D) 56,00,000

55. A factory's output rose from 7,85,000 units to 1 crore 3 lakh units. The increase, rounded to the nearest lakh, is:

- (A) 95,00,000 (B) 96,00,000
(C) 9,50,000 (D) 1,10,85,000

56. A library has 8,40,000 books and plans to double its collection, then round the new total to the nearest ten-lakh for a banner. The banner figure is:

- (A) 10,00,000 (B) 16,80,000
(C) 20,00,000 (D) 17,00,000

57. Three readings are 6,00,060, 6,06,000 and 60,006. By how much does the largest exceed the smallest?

- (A) 5,45,994 (B) 5,99,994
(C) 5,94,000 (D) 6,05,994

58. A bus carries 48 passengers and a city runs 1,250 such buses, each making 6 trips a day. About how many passenger-trips happen daily (round 48 to 50 first)?

- (A) about 37,500 (B) about 3,75,000
(C) about 62,500 (D) about 37,50,000

59. A report states a crowd of 2,49,500 and rounds it to the nearest lakh, while a second report rounds 2,50,500 to the nearest lakh. Which is true?

- (A) both round to 2,00,000 (B) both round to 3,00,000
(C) the first rounds to 3,00,000 and the second to 2,00,000
(D) the first rounds to 2,00,000 and the second to 3,00,000

60. The smallest 10-digit number and the largest 9-digit number differ by:

- (A) 10 (B) 1,00,00,00,000
(C) 0 (D) 1

Solutions — Mathematics (Class 7), Chapter 1: Large Numbers Around Us — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	A	31	D	41	B	51	B
2	B	12	C	22	C	32	A	42	C	52	A
3	D	13	A	23	A	33	A	43	B	53	B
4	B	14	C	24	C	34	D	44	C	54	D
5	B	15	C	25	C	35	A	45	C	55	A
6	D	16	A	26	B	36	A	46	D	56	C
7	D	17	B	27	A	37	C	47	B	57	A
8	B	18	C	28	C	38	A	48	D	58	B
9	B	19	B	29	D	39	D	49	A	59	D
10	A	20	D	30	B	40	D	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Lakhs place has value 1,00,000 and thousands place has value 1,000, so the number is $8 \times 1,00,000 + 8 \times 1,000 = 8,00,000 + 8,000 = 8,08,000$, a 6-digit number. Writing the two 8s side by side as 88,000 is only a 5-digit number and ignores the empty places between them. Putting the second 8 in the hundreds place gives 8,00,800. Reading the first 8 as ten-lakhs gives 80,08,000.

2. (B) Grouped 4,86,30,517: crores 4, ten-lakhs 8, lakhs 6, ten-thousands 3, thousands 0, hundreds 5, tens 1, ones 7. Ten-lakhs 8 plus lakhs 6 is 14. Adding crores and lakhs ($4 + 6$) gives 10; adding lakhs and ten-thousands doubled to $6 + 6$ gives 12; misreading the lakhs digit as 3 (the ten-thousands digit) and adding $6 + 3$ gives 9.

3. (D) Drop commas to get 60700050 (eight digits). Grouping in threes from the right gives 60,700,050. Keeping the Indian commas is no conversion at all; grouping the spare digits on the right as 607,000,50 breaks the 3-3-3 rule; 60,70,0050 uses a 2-2-4 grouping that belongs to neither system.

4. (B) The leftmost digit 9 is in the crores place: place value 9,00,00,000. The 8 is in the lakhs place: place value 8,00,000. The difference is $9,00,00,000 - 8,00,000 = 8,92,00,000$. Subtracting face values ($9 - 8$) gives 1; reading the 8 as ten-lakhs (80,00,000) and subtracting gives 8,20,00,000; mistaking the 8's place value for 80,000 and subtracting gives 8,99,20,000.

5. (B) 3 crore = 3,00,00,000. One ten-thousand = 10,000, so the count is $3,00,00,000 \div 10,000 = 3,000$. Dividing by one lakh instead gives 300; dividing by one thousand gives 30,000; treating crore as ten lakh and dividing wrongly gives 3,00,000.

6. (D) Crores 4, then lakhs part 'four lakh forty thousand' = 4,40,000, then four ones. Building it: 4,04,40,004. Reading 'four lakh' into the ten-lakh slot inflates it to 4,44,40,004; writing 'forty thousand' as 04 thousand gives 4,04,04,004; dropping the empty ten-lakh place collapses it to the 7-digit 44,40,004.

7. (D) 1 million = 10,00,000, so 250 million = $250 \times 10,00,000 = 25,00,00,000$ (twenty-five crore). Treating million as one lakh gives $250 \times 1,00,000 = 2,50,00,000$; treating it as ten thousand gives 25,00,000; adding an extra zero (2,500 lakh as 2,50,00,00,000) is ten times too big.

8. (B) An even number ends in 0 or 2. To be largest, put the biggest digits in the highest places, so try ending in 0: the remaining 9,7,5,2 go in descending order 9,7,5,2 giving 97,520. Ending in 2 forces 0 lower and a smaller leading arrangement (97,5_2 needs 0 inside: $97,502 < 97,520$). 97,250 puts 2 before 5 in the tens/ones region and is smaller; 97,502 ends in 2 and is smaller; 75,920 wastes the lead digit.

9. (B) The 6 sits in the ten-lakhs place, so its place value is 60,00,000 and its face value is 6. Difference $60,00,000 - 6 = 59,99,994$. Subtracting place from face the wrong way still needs the 6 (giving 6 only if you compare face values); reading the 6 as lakhs (6,00,000) and subtracting gives 59,99,940-type slips; forgetting to subtract the face value leaves the place value 60,00,000.

10. (A) From the right: 789, then 56, then 34, then 12 — giving 12,34,56,789 (twelve crore...). 123,456,789 is the International 3-3-3 grouping; 1,23,45,6789 and 1,2345,6789 leave a 4-digit block which the Indian system never allows.

11. (D) Compare from the ten-thousands place: 7,90,000 has 9, 7,09,000 has 0, 7,00,900 has 0. So 7,90,000 is largest. Between the other two, 7,09,000 has 9 thousands while 7,00,900 has 0 thousands, so $7,09,000 > 7,00,900$. Order: $7,90,000 > 7,09,000 > 7,00,900$. The other lists either rank 7,09,000 above 7,90,000 or swap the smallest two.

12. (C) Nearest hundred: the tens digit is 6 (≥ 5), so $7,49,962 \rightarrow 7,50,000$. Rounding that to the nearest thousand leaves 7,50,000 (it is already a multiple of 1,000). Rounding down by truncating gives 7,49,000; stopping at the hundred step alone gives 7,49,900 wrongly

(in fact the hundred-round is already 7,50,000); over-rounding to the nearest lakh gives 8,00,000.

13. (A) 19 rounds to 20, so the estimate is $4,12,000 \times 20 = 82,40,000$. Multiplying by 2 only (forgetting a zero) gives 8,24,000; adding instead of multiplying gives 4,12,019; using the unrounded 19 ($4,12,000 \times 19 = 78,28,000$) is the exact product, not the asked estimate.

14. (C) 3 crore + 5 lakh + 7 thousand + 2 tens = $3,00,00,000 + 5,00,000 + 7,000 + 20 = 3,05,07,020$. Collapsing all the empty places gives 3,57,200; reading the first term as 10 crore gives 30,50,07,020; placing the 5 in the ten-lakh slot gives 3,50,07,020.

15. (C) $9,999 \times 9,999$ is just under $10,000 \times 10,000 = 10,00,00,000$ (a 9-digit number). Exactly, $9,999^2 = 99,980,001 = 9,99,80,001$, which has 8 digits. A product of two 4-digit numbers has 7 or 8 digits, and here the leading digits make it the maximum 8 — not 9, since it stays below 10 crore. The estimate 7 undercounts; 9 wrongly assumes it reaches 10,00,00,000.

16. (A) Five crore = 5,00,00,000; plus five thousand = 5,000; plus five = 5; total 5,00,05,005. Dropping the empty lakh places gives the 6-digit 5,05,005; reading only 'five thousand five' gives 55,005; placing the thousands as ten-thousands gives 5,00,50,005.

17. (B) 30 days $\times$ 4,800 = 1,44,000 copies. Rounding 1,44,000 to the nearest lakh: the ten-thousands digit is 4 (<5), so it rounds down to 1,00,000. Misplacing a zero gives 10,00,000; rounding 1,44,000 down to nothing gives 0; rounding up wrongly gives 2,00,000.

18. (C) Exact need is $9 \times 1,250 = 11,250$ sacks. To never run short you must round UP to a safe round figure, 12,000. Rounding down to 11,000 or 11,200 leaves you short; stocking exactly 11,250 is not a 'round figure' and leaves no margin.

19. (B) One day is $24 \times 60 \times 60 = 86,400$ seconds, so one lakh (1,00,000) seconds is a bit more than one day (about 1 day 4 hours). An hour is only 3,600 seconds (far too few); a week of 6,04,800 seconds is far more than a lakh; a month of about 26 lakh seconds is much more still.

20. (D) 18 lakh = 18,00,000 and 144 crore = 1,44,00,00,000. The fraction is $18,00,000 \div 1,44,00,00,000 = 18 \div 14,400 = 1/800$. Dropping the crore-to-lakh factor of 100 gives $1/8$ (treating crore as lakh) or $1/80$; adding an extra factor of 10 gives $1/8000$.

21. (A) Group 90,90,090: ninety lakh, ninety thousand, ninety. Reading the first 90 as crores gives the wrong 'nine crore...'; reading the middle 90 as 9 thousand gives 'nine thousand'; merging the lakh and ten-thousand blocks gives 'ninety-nine lakh'.

22. (C) 1 crore = 100 lakh, so 12 crore = $12 \times 100 = 1,200$ lakh. Multiplying by 10 instead of 100 gives 120; by 1,000 gives 12,000; forgetting the conversion entirely gives 12.

23. (A) Groups: 408 million, 090 thousand, 007. So 'four hundred eight million ninety thousand seven'. Reading 408 as 480 and 090 as 9 gives the first wrong option; mixing in Indian units (crore/lakh) is the wrong system; reading 090 thousand as 'nine hundred thousand' inflates it.

24. (C) 2 crore 5 lakh = $2,00,00,000 + 5,00,000 = 2,05,00,000 = 20500000$. Grouping in threes from the right: 20,500,000. The option 2,05,00,000 is still Indian commas; 2,500,000 is only 25 lakh (a tenfold error); 205,000,00 breaks the 3-3-3 grouping.

25. (C) Each place to the right is ten times smaller, so two places to the right is $10 \times 10 = 100$ times smaller. Hence the given place value is 100 times larger. Counting one place gives 10; counting the steps instead of the factor gives 2; counting three places gives 1,000.

26. (B) Convert all to plain numbers: 8 crore = 8,00,00,000; 79 million = 7,90,00,000; 7,95,00,000 is itself; 790 lakh = 7,90,00,000. The largest is 8 crore (8,00,00,000), which beats 7,95,00,000 and the two equal 7,90,00,000 values. Mistaking million for crore would wrongly crown 79 million.

27. (A) Rounding to the nearest crore looks at the ten-lakhs (next) digit. Here the part after the crores is 49,52,000, whose leading digit 4 is below 50,00,000, so round DOWN to 5,00,00,000. Rounding up gives 6,00,00,000; rounding to the nearest fifty-lakh is not a place; rounding to the nearest lakh gives 5,49,00,000-type figures, not crore.

28. (C) Round each: $3,87,000 \rightarrow 4,00,000$; $6,12,000 \rightarrow 6,00,000$; $2,49,000 \rightarrow 2,00,000$. Sum $4 + 6 + 2 = 12$ lakh = 12,00,000. Rounding 2,49,000 up to 3,00,000 gives 13,00,000; rounding 3,87,000 down to 3,00,000 gives 11,00,000; the exact sum 12,48,000 is not the rounded estimate asked.

29. (D) $25,000 \times 8,000 = 25 \times 8 \times 1,000 \times 1,000 = 200 \times 10,00,000 = 20,00,00,000$, which has 9 digits. Counting only the eight original trailing structure or losing the carry into a new digit gives 8; further undercounting gives 7; adding a spurious zero gives 10.

30. (B) 50 million = 50,000,000; 500 thousand = 5,00,000; fifty = 50; total 50,500,050. Reading 500 thousand as 5 thousand gives 50,005,050; reading the final 'fifty' as 'five hundred' gives 50,500,500; reading fifty million as five hundred million gives 500,500,050.

31. (D) ₹2 crore 40 lakh = 2,40,00,000. Dividing by 16: $2,40,00,000 \div 16 = 15,00,000$. Misplacing one zero gives ₹1,50,000 or ₹1,50,00,000; dividing only the lakh part gives ₹15,000.

32. (A) A 6-digit palindrome has the form $abccba$. The lakhs digit (1st, 'a') is 4, so the ones digit is also 4: $4bccb4$. The tens digit (2nd from right, 'b') is 7, so the ten-thousands digit is also 7: $47cc74$. The thousands and hundreds digits are the middle pair 'cc'; smallest means $c = 0$, so the thousands digit is 0 (number 470074). Choosing $c = 7$ to match the tens, or $c = 4$ to match the lakhs, or $c = 1$ are larger or unjustified.

33. (A) $60 \text{ lakh} \times 5 = 60,00,000 \times 5 = 3,00,00,000 = 3 \text{ crore}$. So they are equal. Reading $60 \text{ lakh} \times 5$ as 30 lakh wrongly makes 3 crore look bigger; mis-scaling by ten produces the 'ten times' claim.

34. (D) Digit string 7030500 (seven digits). Indian grouping 3-2-2 from the right: 500, then 30, then 70 $\rightarrow$ 70,30,500 (seventy lakh thirty thousand five hundred). The options 7,03,05,00 and 70,30,50,0 mis-split the digits; 7,03,0500 leaves a 4-digit block, which Indian commas forbid.

35. (A) Indian commas fall after 3 digits, then every 2 digits, reading from the right: 890, then 67, then 45, then 23, then 1 — giving 1,23,45,67,890 (one arab twenty-three crore...). The other layouts leave a forbidden 4-digit group (7890) or mix in International 3-3 blocks.

36. (A) The largest place value is the 8 in the crores place: 8,00,00,000. In 8,07,60,400 the digit 6 sits in the ten-thousands place, so its place value is 60,000. The sum is $8,00,00,000 + 60,000 = 8,00,60,000$. Mistaking the 6 for the lakhs digit gives place value 60,00,000 and the sum 8,60,00,000; adding the face values instead gives $8 + 6 = 14$; reading the 6 as the thousands digit gives 6,000 and the sum 8,00,06,000.

37. (C) Convert all: $8.5 \text{ lakh} = 8,50,000$; $0.9 \text{ million} = 9,00,000$; $8,40,000 = 8,40,000$; $95 \text{ ten-thousands} = 95 \times 10,000 = 9,50,000$. Increasing: $8,40,000 < 8,50,000 < 9,00,000 < 9,50,000$. So $8,40,000 < 8.5 \text{ lakh} < 0.9 \text{ million} < 95 \text{ ten-thousands}$. The traps swap 8.5 lakh with 8,40,000 or misplace 0.9 million.

38. (A) Nearest ten-thousand: the thousands digit is 5, so 6,95,000 rounds up to 7,00,000. That value to the nearest lakh stays 7,00,000. Rounding 95 thousand down gives 6,90,000; rounding the lakh step down to 6,00,000 ignores that the ten-thousand step already lifted it past 6,50,000.

39. (D) Round 3,75,000 to the nearest lakh: the ten-thousands digit is 7 (≥ 5), so $\rightarrow$ 4,00,000. A day has 24 hours, so $4,00,000 \times 24 = 96,00,000$ litres. Using 3,00,000 (rounding down) gives about 72 lakh, not offered; using 90,00,000 underestimates; dropping a zero gives 9,00,000; an extra zero gives 9,00,00,000.

40. (D) $7,06,500 = 7 \text{ lakh} + 0 \text{ ten-thousand} + 6 \text{ thousand} + 5 \text{ hundred} + 0 \text{ ten} + 0 \text{ one}$. The first distractor puts 7 in the ten-lakh place (making it 70,00,000+); the second puts 6 in the ten-thousands place (giving 7,65,00... wrong); the third shrinks the 6 and 5 to hundreds and tens (giving 7,00,650).

- 41. (B)** Smallest 4-digit = 1,000; largest 4-digit = 9,999. Product $1,000 \times 9,999 = 99,99,000$, which has 7 digits. Assuming the product reaches 1,00,00,000 (eight digits) overcounts to 8; multiplying digit counts ($4 \times 4 = 16$) is meaningless here; dropping a trailing zero of the 1,000 gives the 6-digit slip.
- 42. (C)** One arab = 1,00,00,00,000 (10-digit). Adding 2 crore (2,00,00,000) and 3 lakh (3,00,000) gives $1,00,00,00,000 + 2,00,00,000 + 3,00,000 = 1,02,03,00,000$. Dropping the arab place gives 1,23,00,000 or 1,02,03,000; reading arab as ten-crore gives 12,03,00,000.
- 43. (B)** $55,000 \times 18 = 9,90,000$. Rounding to the nearest lakh: the ten-thousands digit is 9 (≥ 5), so it rounds UP to 10,00,000. Rounding down gives 9,00,000; misplacing a zero gives 1,00,000 or 1,00,00,000.
- 44. (C)** $45,00,000 \div 4,50,000 = 10$, so the wrong number is 10 times too large (one extra place value). Comparing crore-to-lakh wrongly gives 100; subtracting and dividing carelessly gives 9; over-scaling gives 1,000.
- 45. (C)** 0.6 crore = 60 lakh. Difference 60 lakh – 38 lakh = 22 lakh (22,00,000). Reading the gap in thousands gives 22 thousand; mis-scaling 0.6 crore to 2.8 crore-type errors gives 2.2 crore; adding instead of subtracting gives 98 lakh-type or the wrong 58 lakh.
- 46. (D)** Counting to one lakh needs 1,00,000 numbers; at 2 seconds each that is 2,00,000 seconds. One day = 86,400 seconds, so $2,00,000 \div 86,400 \approx 2.3$ days. Forgetting the 2-second rate gives about 1 day; halving wrongly gives half a day; a week (about 6 lakh seconds) is far too long.
- 47. (B)** Regardless of the value, the Indian system always places its first comma after 3 digits from the right (the hundreds-tens-ones block). So 3 digits come before the first comma. The 2 and 4 answers confuse this with the later 2-digit groups or the International 3-3-3; 1 has no basis.
- 48. (D)** Convert: 9 crore = 9,00,00,000; 88 million = 8,80,00,000; 9,10,00,000 = itself; 900 lakh = 9,00,00,000. So 9,10,00,000 is largest, then 9 crore and 900 lakh tie at 9,00,00,000, then 88 million is smallest. Decreasing: $9,10,00,000 > 9 \text{ crore} = 900 \text{ lakh} > 88 \text{ million}$. Other orders misrank 88 million or 9,10,00,000.
- 49. (A)** Nearest thousand: the hundreds digit is 5, so 9,99,500 rounds up to 10,00,000. That to the nearest lakh stays 10,00,000. Rounding 500 down gives 9,99,000; rounding the lakh step down to 9,00,000 ignores that the value already crossed to ten lakh.
- 50. (C)** 1,985 rounds to 2,000, so the estimate is $5,000 \times 2,000 = 1,00,00,000$. The exact product $5,000 \times 1,985 = 99,25,000$ is not the asked estimate; dropping a zero gives 10,00,000; dropping two zeros gives 1,00,000.

51. (B) $80,07,060 = 8 \text{ ten-lakhs} + 7 \text{ thousands} + 6 \text{ tens}$. The first distractor puts 8 in the lakh place (8,07,060, too small by a place); the second moves 7 to ten-thousands and 6 to hundreds (80,70,600); the third puts 7 in hundreds (80,00,760).

52. (A) $3,00,000 \times 3,00,000 = 9 \times 1,00,000 \times 1,00,000 = 9 \times 10,00,00,00,000 = 90,00,00,00,000$, which has 11 digits. Undercounting the carry gives 10; an extra zero gives 12; counting only one factor's digits gives 6.

53. (B) 1 crore = 1,00,00,000. After draining: $1,00,00,000 - 35,80,000 = 64,20,000$. After pumping back: $64,20,000 + 12,50,000 = 76,70,000$. Stopping after the drain gives 64,20,000; subtracting both amounts gives 51,70,000-type or the 47,30,000 slip; adding both gives 1,18,30,000.

54. (D) Rounding to the nearest ten-lakh, the half-way band that maps to 50,00,000 runs from 45,00,000 up to just below 55,00,000. 45,30,000, 54,90,000 and 50,40,000 all lie in this band. 56,00,000 lies above 55,00,000, so it rounds to 60,00,000 instead — it cannot round to 50,00,000.

55. (A) 1 crore 3 lakh = 1,03,00,000. Increase = $1,03,00,000 - 7,85,000 = 95,15,000$. Rounded to the nearest lakh (ten-thousands digit 1, <5) $\rightarrow 95,00,000$. Rounding 95,15,000 up wrongly gives 96,00,000; misplacing a zero gives 9,50,000; adding instead of subtracting gives 1,10,85,000.

56. (C) Doubling: $8,40,000 \times 2 = 16,80,000$. The two nearest ten-lakh multiples are 10,00,000 and 20,00,000; 16,80,000 is 6,80,000 above the first but only 3,20,000 below the second, so it rounds UP to 20,00,000. The exact total 16,80,000 is not yet rounded; 10,00,000 is the farther multiple; 17,00,000 rounds to the wrong place (lakh, not ten-lakh).

57. (A) Compare: 6,06,000 and 6,00,060 are both about six lakh, while 60,006 is only a 5-digit number near sixty thousand, so it is the smallest; 6,06,000 is the largest. Difference $6,06,000 - 60,006 = 5,45,994$. Subtracting 60 instead of 60,006 gives 5,99,994-type slips; ignoring the small digits gives 5,94,000; subtracting from 6,06,000 the wrong reading gives 6,05,994.

58. (B) Round 48 to 50. Daily passenger-trips $\approx 50 \times 1,250 \times 6 = 50 \times 7,500 = 3,75,000$. Dropping a zero gives 37,500; forgetting the 6 trips ($50 \times 1,250 = 62,500$) gives 62,500; an extra zero gives 37,50,000.

59. (D) For 2,49,500 the ten-thousands digit is 4 (<5), so it rounds down to 2,00,000. For 2,50,500 the ten-thousands digit is 5 (≥ 5), so it rounds up to 3,00,000. They split across the half-lakh boundary, so neither 'both same' option holds, and the directions are not reversed.

60. (D) The smallest 10-digit number is 1,00,00,00,000 and the largest 9-digit number is 99,99,99,999. Consecutive integers differ by $1,00,00,00,000 - 99,99,99,999 = 1$. The factor-of-ten relationship suggests 10 wrongly; the smallest 10-digit number itself is 1,00,00,00,000, not the difference; they are not equal, so not 0.